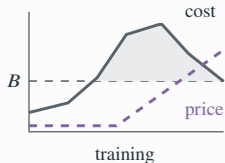
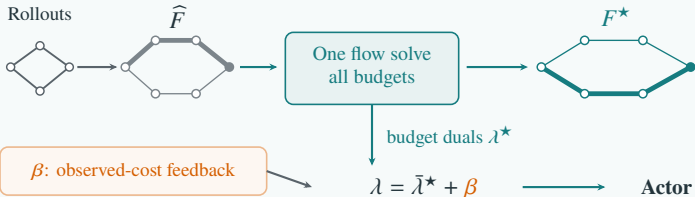


Feedback-only prices react after violations

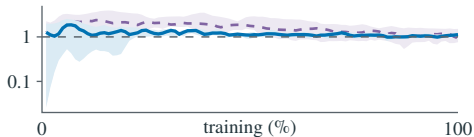


SafeOT-Dual solves the price from rollouts



Cost / budget (log)

SafePO: $0.37 \times V$, 10/10 lower

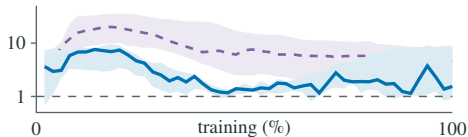


Final ΔR_{norm} : -0.009 $[-0.164, 0.138]$

V = cumulative violation during training

— SafeOT-Dual - - - TRPO-Lag

Bullet: $0.22 \times V$, 6/6 lower



Final ΔR_{norm} : -0.185 $[-0.646, 0.116]$